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A Bayesian analysis, as stressed by Berger (1985, p.120) and Berger and Wolpert (1984) is always conditional on the specific given observations. This approach, which is shared by the authors in their treatment of information, seems more purely Bayesian than even that of Lindley (1956) where the *expected* information over the sample space is studied. Bayesians usually try to avoid such averaging. The subsequent comments will, to a large extent, be concerned with questions concerning this Bayesian approach.

In order to interpret the measure of amount of information provided by the experiment it is necessary to consider the information measure $I = E(\log p(X))$ for a probability density p (cf. Harris (1982), Rosenkrantz (1977), Kendall (1973)). Properties of I show that it can be viewed as a descriptive measure of the distribution p which, in the discrete case, measures the extent to which the probability is concentrated (large values of I) on a few points or dispersed over many points (small values of I). It does not depend on the actual values X can take on, nor on an ordering of them. (If X has a continuous distribution, analogous properties hold). We can therefore say roughly that $I_x = I_{\theta|x} - I_{\theta}$, studied in the paper, measures the difference in dispersion of the probability in the posterior and prior distributions. With this interpretation in mind, we now turn to some unsatisfying features of the measure I_x .

First note that, although $E_X(I_X) \geq 0$ (Lindley (1956)), it may happen that $I_x < 0$ for some x . Thus there can be loss of information through sampling, which is certainly counterintuitive. Let us take a closer look at this. Lindley (1956) states:

"the experimental result may reduce the amount of information. This can happen when a 'surprising' value of x occurs; granted the correctness of the experimental technique, the 'surprise' may result in our being less sure about θ than before the experiment."

In terms of our interpretation of the information measure, this means that the probability is more dispersed in $p(\theta|x)$ than in $p(\theta)$ and $p(\theta|x)$ is flatter than $p(\theta)$. However, a "surprising" value could also lead to *positive* information when the posterior distribution contradicts the prior to a sufficient degree. As a simple example, let X be a Bernoulli random variable with parameter θ . Consider a two point prior with $p(\theta_1) = p_1$, $p(\theta_2) = p_2$, $p_1 < p_2$ and $p_1 + p_2 = 1$. Suppose $x = 0$ is observed (a "bad" observation) and $p(\theta_1|0)$ and $p(\theta_2|0)$ are the posterior probabilities. Then it is easily seen that $p(\theta_1|0) > p(\theta_1)$ (which could mean a flatter posterior) is equivalent to $\theta_1 < \theta_2$, which we may assume to be the case. It can happen, however, that $p(\theta_1|0) > p(\theta_2|0)$, namely when

$$(1) \quad (1 - \theta_1)p_1 > (1 - \theta_2)p_2.$$

Suppose, for instance, that $\theta_1 = 0,25$, $\theta_2 = 0,75$, $p_1 = 0,3$ and $p_2 = 0,7$. Then $p(\theta_1|0) = 0,5625$ and $p(\theta_2|0) = 0,4375$. Also $I_\theta = -0,6109$, while $I_{\theta|x} = -0,6853$ so that $I_x = -0,0744$, a *negative* information.

On the other hand, with $x = 0$ *positive* information could accompany the same conflict between the prior and posterior, namely when the two sides of (1) differ enough. If $\theta_1 = 0,2$ and $\theta_2 = 0,8$ with $p_1 = 0,4$ and $p_2 = 0,6$, we get $p(\theta_1|0) = 0,7273$ and $p(\theta_2|0) = 0,2727$. Then $I_\theta = -0,6730$, $I_{\theta|0} = -0,5860$ so that $I_x = 0,0870$. These examples raise questions about the meaning of negative and positive information.

The phenomena of negative information in an observation and a posterior distribution in conflict with a prior distribution also raise questions about the Bayesian approach to statistics. These questions focus on the idea of a prior distribution.

A possible reason for conflicting prior and posterior distributions is the problem that it is not always clear how much one can trust on the prior. An additional difficulty is the fact that priors are sometimes chosen for mathematical tractability. But the function of the prior is to express in accurate terms the actual prior information which is available. A mathematically convenient prior distribution is not valuable if it misrepresents the true situation (cf. Barnett (1973) p.188).

A practical question that arises is the following: Suppose a prior is specified, observations taken, the information in the observations computed and it turns out to be negative? Already one has a suspicion that something is wrong. What should then be done in practice? Suppose the posterior distribution is computed and it is in conflict with the prior. Where does this lead? (cf. Kempthorne (1972, p.176).)

In view of these questions the use of noninformative priors seem more attractive. A possibility is the use of the so-called maximum entropy priors, which incorporate partial prior information, but are most noninformative on other aspects. Unfortunately this approach seems to have difficulties (cf. Berger (1985) p.90-94). Another unsatisfying aspect is the sensitivity of the information measure to change in the prior distribution as shown in examples in the paper. Perhaps the expected information $E_X(I_X)$ is a more appropriate measure and less vulnerable. Further investigation concerning robustness seems to be called for, perhaps considering contaminated priors as in Berger and Berliner (1986).

Although the present discussion is conducted in a Bayesian framework, it is instructive to contrast this with the frequency view of information. Let us first remark that there is a large variety of measures of information in the literature (cf. Papaioannou (1985)). From a frequency point of view we have "zero information" about the unknown parameter before the experiment. The best known measure of information in this case is the Fisher information, defined as

$$(2) \quad i(\theta) = E_{X|\theta} \left[\frac{\partial \log f(x|\theta)}{\partial \theta} \right]^2$$

where $f(x|\theta)$ is the density function or the probability function of the data X . This measure has a nice intuitive interpretation: The derivative in (2) describes the rate of change of f as the parameter θ changes. With the frequency view there is no concept of information in the specific data, but rather a concept of expected information. Large values of $i(\theta)$ indicate that, on the average, the distribution f is quite sensitive to changes in the parameter θ . The measure $i(\theta)$ is, of course, also connected with various optimality properties of statistical procedures, such as minimum variance unbiased estimation and asymptotic optimality of maximum likelihood estimation.

We see that I_x and $i(\theta)$ have quite different interpretations. The measure I_x is based on differences in the dispersion of the probabilities for θ before and after the experiment. Also it differs for different values of x . On the other hand we see that $i(\theta)$ may be called a measure of the intrinsic accuracy of the distribution, or its sensitivity with respect to the parameter. There is a real difference in the two approaches.

As another difference, note that if $i_n(\theta)$ denotes the Fisher information in n independent observations with distribution f , it is well known that $i_n(\theta) = ni_1(\theta)$. This means that each observation contains the same amount of information and the information in a sample is the sum of the information values in the observations. This is in contrast with the information measure as discussed in the paper, since in that case additional observations can even decrease the information! Again, it seems more plausible to consider the expected information about which Lindley (1956) proved that it never decreases with more observations. (An interesting comment is that by Basu (1958) who says that intuitively one should not expect a second observation to add the same amount of new information as the first. For example, suppose it is known that observations come from either a $N(0, 1)$ or a $N(5, 1)$ distribution. Then the first observation makes clear from which population the sampling is done and a second observation can add very little information which one does not already possess).

The connection between the information measure I_x and sequential analysis is interesting since it indicates a valuable interrelationship between

Bayes theory and the frequency approach. Lindley (1956) suggests continuing sampling until enough information is obtained. If $\delta < 0$ is prescribed, the experimentation continues until the information $I_{x_1, \dots, x_n | \theta} \geq \delta$. Note that here the information in the specific data is used and not the expected information. Lindley shows that the well known Sequential probability ratio test (SPRT) of Wald (1947) arises in this way. De Groot (1962) comes to similar conclusions with much more general information measures of the same character. The SPRT also arises as a Bayes procedure (cf. Lehmann (1959) p. 104-110). The SPRT is a standard procedure in frequency theory, which indicates a connection with the Bayesian approach. The connection is of the following kind: If a Bayes solution can be found for a problem, it often has optimality properties if viewed from the frequency perspective. Thus Bayes theory can be used to suggest procedures and to prove optimality properties from a frequency point of view.

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